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Studierichting: Informatica
Oefeningenreeks 6D nr. 30.2

$$\sum_{n=0}^{\infty} \frac{5^n}{2^n}$$

d'Alembert: $\lim\limits_{n \to +\infty} \frac{x_{n+1}}{x_n} = \lim\limits_{n \to +\infty} \frac{\frac{5^{n+1}}{2^{n+1}}}{\frac{5^n}{2^n}}$

$= \lim\limits_{n \to +\infty} \frac{5^{n+1}}{2^{n+1}} \cdot \frac{2^n}{5^n} = \frac{5}{2} > 1 \Rightarrow$ Divergent

References